

Direct-Route Check of the Updated Shortcut Calculation

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Purpose and conclusion

This note follows the original time-convolution route in the updated manuscript and performs the calculation explicitly. The result is a corrected direct-route version of the long Eq. (41) expansion, together with a check showing that the same expression recompresses to the closed form

$$\boxed{\tilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}} \quad (1)$$

where

$$N = 2L, \quad a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1/z - 1}{q}, \quad U_{-1} = 0.$$

Thus the closed form is not a different model or a shortcut around the calculation. It is the simplified form of the original convolution route after all intermediate poles have been summed.

1 Corrected starting point

Let

$$\lambda = \beta(1-q).$$

When the shortcut is $u \rightarrow v$ and v is the absorbing target, the shortcut is killing in the transient state space. The corrected first-passage generating function is

$$\boxed{\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + qz \tilde{W}_{n_0}(u, z) \left[1 - \tilde{\mathcal{F}}_u(v, z)\right] \tilde{H}(z)} \quad (2)$$

with

$$\tilde{H}(z) = \frac{\beta(1-q)}{q} \frac{1}{1 + z\beta(1-q)\tilde{W}_u(u, z)}. \quad (3)$$

For the antipodal case $|u - v| = N/2$, this is

$$\tilde{H}(z) = \frac{1}{a + \phi(z)}. \quad (4)$$

The sign in (2) is positive. Therefore the corrected time-domain convolution is

$$\boxed{F_{n_0}(v, t) = \mathcal{F}_{n_0}(v, t) + q \sum_{t'=0}^{t-1} W_{n_0}(u, t-1-t') \sum_{t''=0}^{t'} [\delta_{t', t''} - \mathcal{F}_u(v, t' - t'')] H(t'')} \quad (5)$$

for $t \geq 1$. This is the corrected version of the updated manuscript's Eq. (40).

2 Mode expansions used in the direct route

For $N = 2L$, the no-shortcut first-passage probability is

$$\mathcal{F}_{n_0}(v, t) = \sum_{\ell=1}^L f_{\ell}(n_0, v) \alpha_{\ell}^{t-1}, \quad t \geq 1. \quad (6)$$

The killed propagator is

$$W_{n_0}(u, t) = \sum_{r=1}^{N-1} G_r^{(1)}(n_0, u, v) \gamma_r^t + \sum_{r=1}^L g_r^{(2)}(n_0, u, v) \alpha_r^t. \quad (7)$$

Here

$$\alpha_r = 1 - q + q \cos \frac{(2r-1)\pi}{N}, \quad \gamma_r = 1 - q + q \cos \frac{2\pi r}{N}.$$

The corrected H -poles are the roots of the corrected denominator. Write

$$s = 1/z, \quad y_s = 1 + \frac{s-1}{q}.$$

For the antipodal case

$$\tilde{H}(z) = \frac{T_L(y)}{aT_L(y) + U_{L-1}(y)}. \quad (8)$$

Define

$$D(y) = aT_L(y) + U_{L-1}(y).$$

Let

$$h_0 = H(0) = \frac{\beta(1-q)}{q} = \frac{1}{a}.$$

Since $\tilde{H}(z)$ has numerator and denominator of the same degree, its partial-fraction expansion has a constant term. Equivalently, $H(0)$ must be kept separate. If $D(y_k) = 0$ and $s_k = 1 - q + qy_k$, then for $t \geq 1$

$$H(t) = \sum_k c_k s_k^{t-1}, \quad \boxed{c_k = \frac{qT_L(y_k)}{D'(y_k)}}. \quad (9)$$

This residue formula is the safest way to write the corrected version of Eq. (39): it is tied directly to the corrected denominator $D(y)$. In practice one should solve $D(y) = 0$ itself, not only the trigonometric parametrization $y = \cos \theta$. Some roots can have $y < -1$ while still giving an admissible $|s_k| < 1$ pole after $s_k = 1 - q + qy_k$.

3 The convolution kernels

Equation (5) contains two kinds of sums, and each must be split into the $H(0) = h_0$ contribution and the positive-time $H(c)$, $c \geq 1$, pole contribution.

Two-factor convolution

The $H(0)$ part gives simply

$$A_t^{(0)}(x) = x^{t-1}. \quad (10)$$

For a positive-time pole mode s^{c-1} , with $a + c = t - 1$ and $c \geq 1$,

$$\begin{aligned} \sum_{\substack{a+c=t-1 \\ c \geq 1}} x^a s^{c-1} &= \sum_{c=1}^{t-1} x^{t-1-c} s^{c-1} \\ &= \boxed{\frac{s^{t-1} - x^{t-1}}{s - x}}. \end{aligned} \quad (11)$$

Define

$$A_t^{(+)}(x; s) = \frac{s^{t-1} - x^{t-1}}{s - x}. \quad (12)$$

If $x = s$, the expression is understood as the limit

$$A_t^{(+)}(s; s) = (t - 1)s^{t-2}.$$

For $t = 1$, $A_1^{(+)} = 0$, as there is no room for a positive-time H -factor.

Three-factor convolution

The negative part of $1 - \mathcal{F}_u(v, \cdot)$ contributes a first-passage mode y^{b-1} with $b \geq 1$. The $H(0)$ contribution is

$$B_t^{(0)}(x, y) = \sum_{\substack{a+b=t-1 \\ b \geq 1}} x^a y^{b-1} = \frac{x^{t-1} - y^{t-1}}{x - y}. \quad (13)$$

For the positive-time H -pole contribution, $a + b + c = t - 1$, $b \geq 1$, and $c \geq 1$:

$$\sum_{\substack{a+b+c=t-1 \\ b \geq 1, c \geq 1}} x^a y^{b-1} s^{c-1} = [z^{t-3}] \frac{1}{(1 - zx)(1 - zy)(1 - zs)}. \quad (14)$$

Define

$$B_t^{(+)}(x, y; s) = [z^{t-3}] \frac{1}{(1 - zx)(1 - zy)(1 - zs)}. \quad (15)$$

When x, y, s are distinct,

$$B_t^{(+)}(x, y; s) = \frac{x^{t-1}}{(x - y)(x - s)} + \frac{y^{t-1}}{(y - x)(y - s)} + \frac{s^{t-1}}{(s - x)(s - y)}. \quad (16)$$

Coincident cases are obtained by taking limits. This is where terms such as $t\alpha^t$ can appear in an uncompressed expansion. Such terms are not automatically true tail terms; they may cancel after the complete expression is recombined.

4 Corrected direct-route Eq. (41)

Substitute (6), (7), and (9) into (5), keeping the $H(0)$ part separate. The corrected long expression is

$$\begin{aligned}
 F_{n_0}(v, t) = & \sum_{\ell=1}^L f_{\ell}(n_0, v) \alpha_{\ell}^{t-1} \\
 & + q h_0 \left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0, u, v) \gamma_r^{t-1} + \sum_{r=1}^L g_r^{(2)}(n_0, u, v) \alpha_r^{t-1} \right] \\
 & + q \sum_k c_k \left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0, u, v) A_t^{(+)}(\gamma_r; s_k) + \sum_{r=1}^L g_r^{(2)}(n_0, u, v) A_t^{(+)}(\alpha_r; s_k) \right] \\
 & - q h_0 \sum_{\ell=1}^L f_{\ell}(u, v) \left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0, u, v) B_t^{(0)}(\gamma_r, \alpha_{\ell}) \right. \\
 & \quad \left. + \sum_{r=1}^L g_r^{(2)}(n_0, u, v) B_t^{(0)}(\alpha_r, \alpha_{\ell}) \right] \\
 & - q \sum_k c_k \sum_{\ell=1}^L f_{\ell}(u, v) \left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0, u, v) B_t^{(+)}(\gamma_r, \alpha_{\ell}; s_k) \right. \\
 & \quad \left. + \sum_{r=1}^L g_r^{(2)}(n_0, u, v) B_t^{(+)}(\alpha_r, \alpha_{\ell}; s_k) \right].
 \end{aligned} \tag{17}$$

This is the corrected direct-route replacement for the updated manuscript's Eq. (41). Compared with the printed expression, the baseline term must target v , the global sign inherits the positive sign in Eq. (40), and the $g^{(2)}$ modes are paired with α_r , not with γ_r .

5 Why the direct route recompresses to the closed form

Now specialize to $N = 2L$, $v = 0$, $u = L$, and let ρ be the folded distance from the starting site to the absorbing target. The ingredients in (2) have compact Chebyshev forms:

$$\tilde{\mathcal{F}}_{\rho}(0, z) = \frac{T_{L-\rho}(y)}{T_L(y)}, \tag{18}$$

$$\tilde{W}_{\rho}(L, z) = \frac{U_{\rho-1}(y)}{z q T_L(y)}, \tag{19}$$

$$1 - \tilde{\mathcal{F}}_L(0, z) = 1 - \frac{1}{T_L(y)} = \frac{T_L(y) - 1}{T_L(y)}, \tag{20}$$

$$\tilde{H}(z) = \frac{T_L(y)}{a T_L(y) + U_{L-1}(y)}. \tag{21}$$

Substitution into (2) gives

$$\begin{aligned}
 \tilde{F}_{\rho}(z) &= \frac{T_{L-\rho}}{T_L} + q z \frac{U_{\rho-1}}{z q T_L} \frac{T_L - 1}{T_L} \frac{T_L}{a T_L + U_{L-1}} \\
 &= \frac{T_{L-\rho}(a T_L + U_{L-1}) + U_{\rho-1}(T_L - 1)}{T_L(a T_L + U_{L-1})}.
 \end{aligned} \tag{22}$$

The only remaining algebra is a Chebyshev identity:

$$T_L(y) U_{L-\rho-1}(y) = T_{L-\rho}(y) U_{L-1}(y) - U_{\rho-1}(y). \tag{23}$$

Using (23) in (22) gives exactly (1):

$$\tilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}.$$

This proves that the closed form is the direct-route expression after the intermediate α, γ, s terms have been recombined.

6 Tail decay from the recompressed expression

Let

$$N_\rho(y) = aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y), \quad D(y) = aT_L(y) + U_{L-1}(y).$$

The poles of $\tilde{F}_\rho(z)$ are the roots $D(y_k) = 0$. With $s_k = 1 - q + qy_k$, simple roots give

$$F_\rho(t) = \sum_k B_{\rho k} s_k^{t-1}, \quad B_{\rho k} = \frac{q N_\rho(y_k)}{D'(y_k)}. \quad (24)$$

The long-time tail is therefore

$$F_\rho(t) \sim B_{\rho 1} s_1^{t-1}, \quad \frac{F_\rho(t+1)}{F_\rho(t)} \rightarrow s_1. \quad (25)$$

It is not generically $t\alpha_1^t$.

7 Check of the figure in the updated PDF

The figure in the updated PDF is not completely useless. Interpreted narrowly, it is a graphical representation of the poles of H : the intersections of

$$\phi(1/s)$$

with the corrected horizontal line

$$-a = -\frac{q}{\beta(1-q)}.$$

That part is consistent with the corrected denominator $a + \phi(1/s) = 0$.

However, the figure is better treated as a pole-location visualization rather than as a standalone way to infer a finite transition at

$$s_1 = \gamma_1 = 1 - q + q \cos \frac{2\pi}{N}.$$

For the corrected equation,

$$D(\cos \theta) = a \cos(L\theta) + \frac{\sin(L\theta)}{\sin \theta} = 0, \quad s_1 = 1 - q + q \cos \theta_1,$$

and the largest root satisfies

$$\boxed{\gamma_1 < s_1 < \alpha_1} \quad (0 < \beta \leq 1). \quad (26)$$

At $s = \gamma_1$, equivalently $\theta = \pi/L$, the ϕ -term is zero, so the corrected denominator is $a > 0$, not zero. Thus $s_1 = \gamma_1$ is only approached in the infinite-shortcut-strength limit $a \rightarrow 0$, not reached for finite positive admissible β .

There is also a caption typo in the updated PDF: the sentence saying that $q > 1$ is required for a nonzero shortcut should be replaced. The shortcut mass is $\beta(1 - q)$, so one needs $0 < q < 1$ and $\beta > 0$, not $q > 1$.

Suggested replacement caption sentence

The intersections with the horizontal line $-q/[\beta(1-q)]$ give the poles s_k of the corrected auxiliary function $\tilde{H}_{N,N/2}(z) = 1/(q/[\beta(1-q)] + \phi(z))$. This figure is a visualization of the H -poles only; it does not by itself determine the finite-time one-peak/two-peak transition. For finite $0 < \beta \leq 1$, the dominant corrected pole satisfies $\gamma_1 < s_1 < \alpha_1$, so the proposed finite crossing $s_1 = \gamma_1$ does not occur.

8 Validation summary

Check	Result
Corrected plus/plus first-passage formula vs finite stochastic matrix	2.08×10^{-17} max difference
Corrected direct-route Eq. (17) vs finite stochastic matrix	1.32×10^{-16} max difference
Updated Eq. (32) denominator sign vs finite stochastic matrix	3.29×10^{-3} max difference
Updated Eq. (33) minus prefactor vs finite stochastic matrix	6.34×10^{-2} max difference
Closed form (1) vs finite stochastic matrix	7.63×10^{-17} max difference

For $N = 100$, $q = 2/3$, and the fixed shortcut $u = 6 \rightarrow v = 56$, the clear visual double-peak scan finds clear double peaks up to about $\beta = 0.03$ for $n_0 = 1, 2, 3$, and no clear double peak at $\beta_c = 2q/[(1-q)N] = 0.04$. This supports treating bimodality as a finite-time shape question, not as a direct consequence of the asymptotic tail crossing.

Deliverable recommendation

One useful package to send back is this direct-route note together with the short benchmark note. The direct-route note shows how the corrected long convolution is obtained and why it recompresses to the closed form; the benchmark note gives the numerical evidence for updating the remaining signs in the displayed first-passage formulae.